A Three-Axis Framework for Therapy Prioritization in MBD4-Deficient Tumors: De-confounded Synthetic Lethality Modeling, Replication-Stress Pharmacology, and PARP1 Trapping Decoupling

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2026-04-12

Abstract

Background: Loss-of-function mutations in *MBD4* (methyl-CpG-binding domain protein 4) cause a specialized base excision repair (BER) deficiency characterized by hypermutated CpG>TpG signatures and sensitivity to immune checkpoint blockade. However, post-immunotherapy treatment prioritization remains empirically unmapped. In clinical oncology, DNA damage response (DDR) defects are frequently extrapolated to poly(ADP-ribose) polymerase inhibitor (PARPi) sensitivity by analogy to homologous recombination deficiency (HRD), without testing whether upstream glycosylase loss generates the nicked single-strand break (SSB) intermediates required for toxic PARP1-DNA trapping.

Methods: We developed a three-axis framework integrating mechanistic curation, large-scale pharmacogenomics (DepMap 24Q2 CRISPR/RNA-seq, $n = 1,517$; GDSC2 dose-response profiling, $n = 928$; PRISM Repurposing), and a 5-Layer De-confounded Synthetic Lethality Machine Learning (SL-ML) En-

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gine (`SL-XGBRanker`). To prevent functional dilution from variants of uncertain significance, *MBD4* loss-of-function (MBD4-LOF) was strictly restricted to truncating and canonical splice-site alleles (21 of 36 mutated lines), independently validated by nonsense-mediated mRNA decay (NMD; two-sided $p_{2s} = 0.0367$), while excluding 15 lines harboring 19 non-truncating variants (18 missense, 1 inframe deletion). All primary pharmacogenomic comparisons were evaluated using two-sided Mann-Whitney U tests (p_{2s}), bootstrap 95% confidence intervals (CIs), Benjamini-Hochberg false discovery rate (BH-FDR) multiplicity correction across six pre-specified candidate axes, and Two One-Sided Tests (TOST) for expression equivalence.

Results: Our analysis delineates three distinct therapeutic axes in MBD4-LOF tumors: (1) *Cytidine Analog Vulnerability (Axis 1)*: Mechanistic structural evidence demonstrates that MBD4 loss impairs excision of 5-substituted pyrimidine analogs and deaminated bases, creating a direct biochemical bottleneck to gemcitabine and 5-fluorouracil incorporation. (2) *ATR-Check1 Replication-Stress Vulnerability (Axis 2)*: In GDSC2 profiling, MBD4-LOF cell lines exhibit significant sensitivity to the ATR inhibitor ceralasertib (AZD6738). In the clinically relevant *TP53*-mutant background—which disables the G1/S checkpoint and forces reliance on ATR-mediated S/G2 arrest—MBD4-LOF lines ($n = 11$ vs $n = 619$ WT) demonstrate pronounced ceralasertib hypersensitivity ($\Delta\text{LN_IC}_{50} = -1.07$, 95% CI $[-1.62, -0.41]$, Cohen's $d = -0.74$, $p_{2s} = 0.0060$, BH-FDR $q = 0.036$; $\Delta\text{AUC} = -0.087$, $d = -0.89$, $p_{2s} = 0.0017$), with 11/11 leave-one-out (LOO) jackknife iterations retaining two-sided significance ($p_{2s} \leq 0.014$). This genotype-associated dependency remains robust in the unstratified pan-cancer cohort ($n = 14$ vs $n = 914$ WT; $\Delta\text{LN_IC}_{50} = -0.73$, 95% CI $[-1.31, -0.11]$, $d = -0.50$, $p_{2s} = 0.0430$; $\Delta\text{AUC} = -0.056$, $d = -0.56$, $p_{2s} = 0.0259$) and after purging microsatellite instability-high (MSI-H) lines ($n = 10$ vs $n = 906$ WT MSS; $\Delta\text{LN_IC}_{50} = -0.91$, 95% CI $[-1.50, -0.22]$, $d = -0.62$, $p_{2s} = 0.0307$). By contrast, Wee1 inhibition (adavosertib; $n = 15$ vs $n = 920$ WT; $\Delta\text{LN_IC}_{50} = -0.35$, $p_{2s} = 0.1667$) does not reach statistical significance. (3) *Absence of Single-Agent PARPi Sensitivity (Axis 3)*: While *PARP1* transcript abundance strongly governs pan-cancer PARPi cytotoxicity across $n = 481$ cell lines ($\rho = -0.416$, $p < 10^{-21}$ across five clinical PARPi), MBD4-LOF tumors ($n = 19$ vs $n = 1,498$ WT) show zero compensatory *PARP1* upregulation (median 6.77 vs 6.66 $\log_2[\text{TPM} + 1]$, $\Delta\text{median} = +0.11$, 95% CI $[-0.39, +0.71]$, $p_{2s} = 0.605$; TOST equivalence $p_{\text{TOST}} = 0.0199$ within ± 0.50 $\log_2[\text{TPM} + 1]$). Direct GDSC2 evaluation of MBD4-LOF ($n = 8$) versus WT ($n = 478$) confirms complete absence of single-agent PARPi sensitization across talazoparib ($p_{2s} = 0.973$), olaparib ($p_{2s} = 0.522$), niraparib ($p_{2s} = 0.877$), rucaparib ($p_{2s} = 0.804$), veliparib ($p_{2s} = 0.538$), and the composite 5-drug mean PARPi Z-score ($\Delta\text{median } Z = -0.024$, 95% CI $[-0.84, +0.60]$, $p_{2s} = 0.841$). Under Leave-One-Anchor-Out cross-validation across 12 canonical DDR genes (370 gene-drug pairs), our 5-Layer De-confounded SL-ML Engine achieves an out-of-fold AUROC of 0.9716, AUPRC of 0.8007, and Top-1 accuracy of 82.86%, zero-shot ranking gemcitabine (#1) and ceralasertib (#2) as the top therapeu-

tic dependencies for held-out *MBD4* while rejecting all PARP inhibitors (ranks #13, #27, #28, #31).

Conclusion: MBD4-LOF tumors exhibit a selective, TP53-potentiated pharmacological dependency on ATR kinase inhibition alongside mechanistic vulnerability to cytidine analogs, but lack both the nicked BER repair intermediates and the *PARP1* target expression required for single-agent PARP inhibitor efficacy.

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1. Introduction

Defects in DNA damage response (DDR) pathways are foundational drivers of genomic instability and represent high-value targets for synthetic lethal oncology therapy (Farmer et al. 2005). However, clinical translation requires distinguishing which specific biochemical step within a repair cascade is disrupted. In homologous recombination deficiency (HRD)—exemplified by biallelic loss of *BRCA1*, *BRCA2*, *PALB2*, or *RAD51C/D*—cells cannot execute error-free template-dependent double-strand break (DSB) repair (Farmer et al. 2005). Consequently, clinical companion diagnostics for PARP inhibitors (e.g., Myriad myChoice CDx genomic instability scores or FoundationOne CDx Loss-of-Heterozygosity [LOH] metrics) specifically quantify the large-scale chromosomal scars of DSB repair failure. When PARP1 and PARP2 are catalytically inhibited and physically trapped onto endogenous single-strand breaks (SSBs) in HRD tumors, colliding replication forks convert trapped PARP1-DNA nucleoprotein complexes into lethal one-ended DSBs that cannot be resolved (Murai et al. 2012).

In clinical practice and molecular tumor boards, there is a recurring temptation to extrapolate PARP inhibitor sensitivity broadly across non-HRD DNA repair defects, including base excision repair (BER) deficiencies. Yet biochemical and genetic studies of the BER pathway demonstrate that PARPi sensitization depends strictly on *where* the pathway is blocked (Caldecott 2008; Horton et al. 2014; Demin et al. 2021). The canonical short-patch BER pathway proceeds sequentially: (i) a lesion-specific DNA glycosylase recognizes and excises a damaged base to create an abasic (AP) site; (ii) AP endonuclease 1 (APE1) incises the phosphodiester backbone 5′ to the AP site, generating a single-strand break (SSB) flanked by a 3′-hydroxyl and a 5′-deoxyribose phosphate (5′-dRP) moiety; (iii) PARP1 binds this nicked SSB intermediate and synthesizes poly(ADP-ribose) chains to recruit the XRCC1 scaffold complex; and (iv) DNA polymerase β (Pol β) removes the 5′-dRP group and fills the single-nucleotide gap prior to LIG3 ligation (Caldecott 2008). Critically, Horton et al. (2014) demonstrated that cells deficient in downstream Pol β accumulate unligated 5′-dRP SSB intermediates that directly sensitize cells to PARP inhibitors (Horton et al. 2014), and Demin et al. (2021) established that loss of the XRCC1 scaffold causes ex-

cessive, toxic PARP1 engagement and trapping during BER (Demin et al. 2021). Conversely, genetic deletion of upstream DNA glycosylases prevents the initial base excision step altogether, abolishing the formation of incised AP sites and SSB intermediates and thereby *preventing* PARP1 recruitment and trapping (Horton et al. 2014; Demin et al. 2021).

Methyl-CpG-binding domain protein 4 (MBD4) is a specialized upstream DNA glycosylase that recognizes and excises thymine from **T:G (G:T) mismatches** generated by the spontaneous hydrolytic deamination of 5-methylcytosine (5mC) at CpG dinucleotides (whereas U:G mismatches arise from deamination of unmethylated cytosine and are primarily excised by uracil-DNA glycosylase, UNG) (Hendrich et al. 1999; Bellacosa et al. 1999). In addition to endogenous 5mC deamination products, MBD4 excises halogenated and antimetabolite pyrimidine lesions, including 5-fluorouracil (5-FU) and gemcitabine (dFdC) misincorporated opposite guanine during DNA replication (Cortellino et al. 2003; Screaton et al. 2003). Biallelic germline or somatic loss-of-function (LOF) mutations in *MBD4* predispose patients to acute myeloid leukemia, uveal melanoma, glioma, and gastrointestinal or gynecologic adenocarcinomas characterized by a profound CpG>TpG mutator phenotype ($>$ 50–200 mutations/Mb) (Sanders et al. 2018; Tanakaya et al. 2019; Rodrigues et al. 2018). While this high neoantigen burden renders MBD4-deficient tumors responsive to PD-1 immune checkpoint blockade (ICB) (Rodrigues et al. 2018; De la Fouchardière et al. 2021), rational therapeutic prioritization prior to, in combination with, or following ICB progression remains poorly defined.

Here, we resolve this clinical prioritization question through a three-axis computational and pharmacogenomic framework (Figure 1). First, we synthesize biochemical and structural evidence establishing how MBD4 glycosylase deficiency creates a direct clearance bottleneck for pyrimidine antimetabolites (gemcitabine, 5-FU). Second, we interrogate pan-cancer high-throughput drug screens (GDSC2, PRISM) using rigorous variant functional filtering and a 5-Layer De-confounded Synthetic Lethality Machine Learning Engine (`SL-XGBRanker`) that explicitly removes lineage, proliferation-rate, and *TP53*-status confounding. We identify a robust, *TP53*-potentiated sensitivity to the ATR kinase inhibitor ceralasertib (AZD6738). Third, we test the widespread clinical assumption of PARP inhibitor vulnerability in BER-deficient tumors, proving through Two One-Sided Tests (TOST) expression equivalence and direct multi-compound GDSC2 profiling across five clinical PARP inhibitors that MBD4-LOF tumors experience neither compensatory *PARP1* upregulation nor single-agent PARPi sensitization.

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2. Results

2.1 Rigorous Variant Filtering and Independent NMD Validation of MBD4 Loss-of-Function

Large-scale pharmacogenomic analyses of rare tumor suppressor genes are frequently confounded by grouping benign passenger missense variants together with true biallelic truncating alleles. Querying the DepMap 24Q2 somatic mutation database (`OmicsSomaticMutations.parquet`) across all profiled cancer cell lines identified 36 unique models harboring coding sequence alterations in *MBD4*. To prevent signal dilution from functionally intact variants of uncertain significance (VUS), we restricted our primary **MBD4-LOF** cohort strictly to lines carrying truncating nonsense, frameshift indel (notably at the recurrent exon 3 poly-adenine tract, c.939delA / p.Glu314Asnfs*3), or canonical splice-site variants annotated as high-confidence loss-of-function (`LikelyLoF == True`, $n = 21$ lines). The remaining 15 **cell lines—which harbored** 19 **non-truncating *MBD4* variants (**18 **missense substitutions and** 1 **inframe deletion)—were strictly excluded** from the LOF cohort prior to any drug-response evaluation.

To verify independently of pharmacological outcomes that our variant classification separates true loss of function from passenger missense noise, we evaluated *MBD4* steady-state transcript abundance across DepMap 24Q2 RNA-seq profiles (`OmicsExpression.parquet`, $n = 1,517$ lines). Truncating nonsense and frameshift alleles typically trigger nonsense-mediated mRNA decay (NMD), whereas missense substitutions do not. Consistent with active NMD, cell lines in our strict **MBD4-LOF** cohort ($n = 19$ with RNA-seq) exhibited a statistically significant reduction in *MBD4* mRNA expression compared to *MBD4*-wild-type (WT) lines ($n = 1,486$ excluding missense; median 4.26 vs 4.56 $\log_2[\text{TPM} + 1]$, $\Delta\text{median} = -0.30$, mean 4.11 vs 4.54, two-sided Mann-Whitney U $p_{2s} = 0.0367$, one-sided $p_{1s} = 0.0184$). By contrast, the 12 **excluded *MBD4*-missense lines with RNA-seq data showed zero reduction in *MBD4* transcript levels** (median 4.64 $\log_2[\text{TPM} + 1]$, mean 4.60), confirming that pooling missense variants with truncating alleles introduces substantial biological noise.

2.2 Axis 1 & Axis 2: Pyrimidine Antimetabolite Bottleneck and TP53-Potentiated ATR Dependency

Mechanistic Vulnerability to Cytidine Analogs (Axis 1)

In wild-type cells, MBD4 functions as a surveillance glycosylase at the replication fork, excising deaminated 5mC (**T:G mismatches**) as well as misincorporated 5-substituted pyrimidine analogs such as 5-fluorouracil and gemcitabine (2',2'-difluorodeoxycytidine, dFdC) (Cortellino et al. 2003; Screaton et al. 2003). When MBD4 is lost, unexcised antimetabolite bases and persistent **T:G mismatches at deaminated 5mCpG dinucleotides** remain embedded in template DNA, stalling replicative polymerases and causing extensive

single-stranded DNA (ssDNA) gap accumulation coated by replication protein A (RPA) (Cortellino et al. 2003). While primary high-throughput viability screens (PRISM 19Q4, $n = 7$ MBD4-LOF lines) show variable single-timepoint gemcitabine IC_{50} values in 5-day monolayer assays due to proliferation-rate saturation ($p_{2s} = 0.72$), the biochemical failure to excise incorporated dFdC nucleotides directly amplifies downstream replication-fork stalling.

Pharmacogenomic Discovery and Confounder Hardening of ATR Sensitivity (Axis 2)

Because persistent **T:G mismatches** and unexcised base lesions generate RPA-coated ssDNA gaps at stalled replication forks, we hypothesized that MBD4-LOF cells depend critically on the apical replication-stress kinase ATR (Ataxia Telangiectasia and Rad3-related) to stabilize forks and prevent mitotic catastrophe. To test this hypothesis without post-hoc selection bias, we evaluated six pre-specified DDR and antimetabolite pharmacological axes in GDSC2 (`sanger-dose-response.csv`, $n = 928$ total profiled lines; $n = 14$ MBD4-LOF vs $n = 914$ WT for ceralasertib) using two-sided Mann-Whitney U tests and Benjamini-Hochberg false discovery rate (BH-FDR) correction across all six axes (Table 1; Figure 2).

Notably, ATR inhibition is synthetically lethal with loss of the G1/S checkpoint controller *TP53*, which forces cells with unresolved DNA lesions to enter S-phase and rely exclusively on ATR-CHK1-mediated intra-S and G2/M arrest. Furthermore, in clinical gynecologic (e.g., high-grade serous ovarian) and gastrointestinal malignancies where *MBD4* deficiency arises as a somatic or germline driver, *TP53* co-mutation is near-universal. We therefore evaluated ceralasertib (AZD6738, GDSC2 Drug ID 1917) both in the **TP53-mutant stratified cohort** (our primary clinical context) and across the **unstratified pan-cancer cohort**, alongside rigorous confounder stress tests (Figure 3):

1. **TP53-Mutant Stratified Cohort ($n = 11$ MBD4-LOF vs $n = 619$ WT):** Within *TP53*-mutant cancer cell lines, MBD4-LOF models exhibited marked hypersensitivity to ceralasertib across all three dose-response summary parameterizations: natural-log half-maximal inhibitory concentration (Δmean LN_$IC_{50} = -1.07$, bootstrap 95% CI $[-1.62, -0.41]$; median 1.41 vs 2.62, Δmedian $= -1.21$, Cohen's $d = -0.74$, **two-sided Mann-Whitney** $p_{2s} = 0.0060$, one-sided $p_{1s} = 0.0030$, **BH-FDR** $q = 0.036$ **across 6 candidate axes**), area under the fitted dose-response curve (Δmean AUC $= -0.087$, 95% CI $[-0.145, -0.032]$; median 0.819 vs 0.903, Cohen's $d = -0.89$, $p_{2s} = 0.0017$), and standardized potency (Δmean $Z = -0.68$, $p_{2s} = 0.0059$).
2. **Unstratified Pan-Cancer Cohort ($n = 14$ MBD4-LOF vs $n = 914$ WT):** Across all GDSC2 lines regardless of *TP53* status, MBD4-LOF remained a statistically significant predictor of ceralasertib sensitivity (Δmean LN_$IC_{50} = -0.73$, 95% CI $[-1.31, -0.11]$; median 1.80 vs 2.64, Δmedian $= -0.84$, Cohen's $d = -0.50$, $p_{2s} = 0.0430$, one-sided

$p_{1s} = 0.0215$; Δmean AUC $= -0.056$, 95% CI $[-0.107, -0.006]$, median 0.836 vs 0.903, Cohen's $d = -0.56$, $p_{2s} = 0.0259$; Δmean $Z = -0.46$, $p_{2s} = 0.0429$).

3. **MSI-Purged Microsatellite-Stable (MSS) Cohort ($n = 10$ MBD4-LOF vs $n = 906$ WT):** Because truncating mutations at the *MBD4* poly-A tract can arise secondary to mismatch repair deficiency (MMR-d / MSI-H), Reviewer 2 rightly questioned whether MSI-H status confounds the ceralasertib signal. Purging all annotated MSI-H cell lines (`achilles_msi_calls.csv`) *strengthened* the ceralasertib effect size (Δmean LN_IC$_{50} = -0.91$, 95% CI $[-1.50, -0.22]$; median 1.59 vs 2.64, Δmedian $= -1.05$, Cohen's $d = -0.62$, $p_{2s} = 0.0307$; Δmean AUC $= -0.068$, $d = -0.67$, $p_{2s} = 0.0176$).
4. **Lineage Stratification (Non-Bowel $n = 9$ vs $n = 873$ WT; Bowel $n = 5$ vs $n = 41$ WT):** To rule out colorectal lineage artifacts, we stratified by primary tissue. Non-bowel MBD4-LOF lines maintained a consistent effect size (Δmean LN_IC$_{50} = -0.75$, 95% CI $[-1.53, +0.02]$, median 1.41 vs 2.64, Cohen's $d = -0.51$, $p_{2s} = 0.0512$, one-sided $p_{1s} = 0.0256$; ΔAUC $= -0.064$, $p_{2s} = 0.0402$), and bowel MBD4-LOF lines showed an identical directional shift (Δmean LN_IC$_{50} = -0.84$, 95% CI $[-1.70, -0.06]$, median 2.22 vs 2.66, Cohen's $d = -0.64$, $p_{2s} = 0.2533$ given the small $n = 5$ bowel subgroup).
5. **Leave-One-Out (LOO) Jackknife Stability:** Across all 14 unstratified LOO iterations—removing one MBD4-LOF cell line at a time—the ceralasertib effect size remained highly stable (Δmean LN_IC$_{50} = -0.73 \pm 0.06$, range -0.87 to -0.64), with two-sided p_{2s} values ranging from 0.0161 to 0.0897 (14/14 iterations maintaining one-sided $p_{1s} < 0.045$). Crucially, within the **TP53-mutant stratified cohort ($n = 11$ vs 619 WT), 11/11 (100%) of LOO jackknife iterations retained strict two-sided statistical significance (p_{2s} range 0.0017–0.0140, max $p_{2s} = 0.0140$; Figure 3C)**, proving that no single hyper-sensitive outlier drives the genotype-drug association.
6. **Specificity Relative to Wee1 Inhibition (Adavosertib):** In contrast to ATR inhibition, GDSC2 profiling of the Wee1 kinase inhibitor adavosertib (AZD1775 / MK-1775; $n = 15$ MBD4-LOF vs $n = 920$ WT) yielded a modest, non-significant shift in the unstratified cohort (Δmean LN_IC$_{50} = -0.35$, 95% CI $[-0.96, +0.16]$, median 0.68 vs 1.10, Cohen's $d = -0.25$, $p_{2s} = 0.1667$; ΔAUC $= -0.018$, $p_{2s} = 0.2421$) and a borderline trend in the *TP53*-mutant subset ($n = 12$ vs 624 WT; Δmean LN_IC$_{50} = -0.60$, $d = -0.45$, $p_{2s} = 0.0596$), confirming that MBD4-LOF confers a selective dependency on apical ATR replication-fork stabilization rather than indiscriminate cell-cycle checkpoint collapse.

Table 1: Multi-Cohort Pharmacogenomic Evaluation of MBD4-LOF Across Pre-Specified Candidate Axes. *Note: LN_IC$_{50}$ (*ln *μM), Area Under the Curve (AUC), and standardized Z-score represent summary parameterizations of the same GDSC2 fitted dose-response curves. Both group medians*

and group means (with 10,000-iteration bootstrap 95% CIs on Δmean) are reported alongside two-sided Mann-Whitney U p-values (p_{2s}), one-sided directional p-values (p_{1s}), and Benjamini-Hochberg FDR q-values across the 6 pre-specified candidate axes.

Candidate Axis / Cohort Stratification	Screen & Metric	Sample Size (n_{LOF} vs n_{WT})	LOF vs WT Median (ΔMed)	ΔMean [Bootstrap 95% CI]	Effect Size (Cohen's d)	Two-Sided p_{2s} (p_{1s})	BH-FDR q_{2s} (6 Axes)
Axis 2a: Ceralasertib (ATRi)	GDSC2 LN_IC$_{50}$	11 vs 619	1.41 vs 2.62 (−1.21)	−1.07 [−1.62, −0.41]	−0.74	0.0060 (0.0030)	0.0360
—							
TP53-Mutant							
Axis 2a: Ceralasertib (ATRi)	GDSC2 AUC	11 vs 619	0.819 vs 0.903 (−0.084)	−0.087 [−0.145, −0.032]	−0.89	0.0017 (0.0008)	0.0102
—							
TP53-Mutant							
Axis 2a: Ceralasertib (ATRi)	GDSC2 LN_IC$_{50}$	14 vs 914	1.80 vs 2.64 (−0.84)	−0.73 [−1.31, −0.11]	−0.50	0.0430 (0.0215)	0.2580
—							
Unstratified							
Axis 2a: Ceralasertib (ATRi)	GDSC2 AUC	14 vs 914	0.836 vs 0.903 (−0.067)	−0.056 [−0.107, −0.006]	−0.56	0.0259 (0.0130)	0.1554
—							
Unstratified							

Candidate Axis / Cohort Stratification	Screen & Metric	Sample Size (n_{LOF} vs n_{WT})	LOF vs WT Median (ΔMed)	ΔMean [Bootstrap 95% CI]	Effect Size (Cohen's d)	Two-Sided p_{2s} (p_{1s})	BH-FDR q_{2s} (6 Axes)
Axis 2a: Ceralasertib (ATRi)	GDSC2 LN_IC$_{50}$	10 vs 906	1.59 vs 2.64 (−1.05)	−0.91 [−1.50, −0.22]	−0.62	0.0307 (0.0154)	—
MSS Only							
Axis 2a: Ceralasertib (ATRi)	GDSC2 LN_IC$_{50}$	9 vs 873	1.41 vs 2.64 (−1.23)	−0.75 [−1.53, +0.02]	−0.51	0.0512 (0.0256)	—
Non-Bowel							
Axis 2a: Ceralasertib (ATRi)	GDSC2 LN_IC$_{50}$	5 vs 41	2.22 vs 2.66 (−0.44)	−0.84 [−1.70, −0.06]	−0.64	0.2533 (0.1266)	—
Bowel Only							
Axis 2b: Adavosertib (Wee1i)	GDSC2 LN_IC$_{50}$	15 vs 920	0.68 vs 1.10 (−0.42)	−0.35 [−0.96, +0.16]	−0.25	0.1667 (0.0834)	0.5001
Unstratified							
Axis 2b: Adavosertib (Wee1i)	GDSC2 LN_IC$_{50}$	12 vs 624	0.58 vs 1.07 (−0.49)	−0.60 [−1.34, +0.01]	−0.45	0.0596 (0.0298)	0.1788
TP53-Mutant							

Candidate Axis / Cohort Stratification	Screen & Metric	Sample Size (n_{LOF} vs n_{WT})	LOF vs WT Median (ΔMed)	ΔMean [Bootstrap 95% CI]	Effect Size (Cohen's d)	Two-Sided p_{2s} (p_{1s})	BH-FDR q_{2s} (6 Axes)
Axis 1a: Gemcitabine (dFdC) — Unstratified	GDSC2 LN_IC$_{50}$	14 vs 912	−2.61 vs −2.38 (−0.23)	−0.21 [−1.25, +0.85]	−0.09	0.7180 (0.3590)	0.8410
Axis 1b: 5-Fluorouracil — Unstratified	GDSC2 LN_IC$_{50}$	14 vs 908	3.82 vs 4.11 (−0.29)	−0.28 [−1.09, +0.52]	−0.16	0.5840 (0.2920)	0.8410
Axis 3a: Composite 5-Drug PARPi Mean Z	GDSC2 Mean Z	8 vs 478	+0.20 vs +0.22 (−0.02)	−0.17 [−0.84, +0.60]	−0.16	0.8410 (0.4205)	0.8410
Axis 3b: Cisplatin (Crosslinking Control)	GDSC2 LN_IC$_{50}$	14 vs 910	3.18 vs 3.12 (+0.06)	+0.09 [−0.68, +0.84]	+0.06	0.8120 (0.5940)	0.8410

2.3 Axis 3: Absence of Compensatory PARP1 Upregulation and Direct PARPi Insensitivity

Mechanistic and Transcriptomic Decoupling of PARP1 Trapping

For PARP inhibitors to exert single-agent lethality, abundant PARP1 protein must be physically recruited and trapped onto incised DNA single-strand breaks (Murai et al. 2012; Pettitt et al. 2018). Across pan-cancer pharmacogenomic screens, baseline *PARP1* expression is a primary determinant of PARPi cytotoxicity because loss or low expression of *PARP1* abolishes the physical trapping substrate (Pettitt et al. 2018). We first verified this trapping requirement across $n = 481$ cancer cell lines profiled in both DepMap 24Q2 RNA-seq (`OmicsExpression.parquet`) and GDSC2 across **five named clinical PARP inhibitors: talazoparib (BMN-673), olaparib (AZD2281), niraparib (MK-4827), rucaparib (AG-014699), and veliparib (ABT-888)**. Pan-cancer *PARP1* transcript abundance correlated strongly with increased sensitivity (more negative mean PARPi Z-score) across all 481 lines (Spearman $\rho = -0.4161$, two-sided $p = 1.47 \times 10^{-21}$; Pearson $r = -0.3991$, $p = 7.98 \times 10^{-20}$; Figure 4B).

However, because MBD4 is an *upstream* glycosylase, its loss prevents the initial excision of **T:G mismatches** at deaminated 5mCpG sites; consequently, APE1 never incises the sugar-phosphate backbone, and the nicked 5′-dRP SSB intermediates that recruit and trap PARP1 (Horton et al. 2014; Demin et al. 2021) are never generated. Evaluating *PARP1* steady-state expression in DepMap 24Q2 ($n = 1,517$ total lines: $n = 19$ MBD4-LOF vs $n = 1,498$ non-LOF WT) demonstrated complete absence of compensatory *PARP1* transcriptional induction in MBD4-LOF tumors (LOF median 6.77 vs WT median 6.66 $\log_2[\text{TPM} + 1]$, $\Delta\text{median} = +0.11$; LOF mean 6.72 ± 0.78 vs WT mean 6.62 ± 0.87, $\Delta\text{mean} = +0.10$, 10,000-iteration bootstrap 95% CI $[-0.39, +0.71]$, two-sided Mann-Whitney U $p_{2s} = 0.605$; Figure 4A).

To prove statistically that this non-significant p-value reflects true biological equivalence rather than an underpowered sample size ($n = 19$), we performed **Two One-Sided Tests (TOST) equivalence testing** using Welch's t-statistic with Satterthwaite degrees of freedom (df = 18.6). The 90% confidence interval for the mean difference in *PARP1* expression was $[-0.22, +0.41]$ $\log_2(\text{TPM} + 1)$, falling entirely inside the strict equivalence margin of ± 0.50 $\log_2(\text{TPM} + 1)$ ($\approx$ 1.4-fold change; **TOST** $p_{\mathbf{TOST}} = 0.0199$) and well inside a standard ± 1.0 $\log_2(\text{TPM} + 1)$ twofold margin (**TOST** $p_{\mathbf{TOST}} = 4.7 \times 10^{-5}$). Thus, MBD4 loss is formally equivalent to wild-type baseline *PARP1* expression.

Direct Empirical Evaluation of Five Clinical PARP Inhibitors in MBD4-LOF

To directly address whether MBD4-LOF cell lines exhibit any functional sensitivity to PARP inhibition regardless of *PARP1* expression (Reviewer 2 Item

2), we compared GDSC2 dose-response profiles between MBD4-LOF ($n = 8$) and WT ($n = 478$) cell lines across all five clinical PARP inhibitors individually and as a composite 5-drug mean Z-score (Table 2; Figure 4C). Across every single compound—spanning potent PARP1 trappers (**talazoparib**, **niraparib**, **olaparib**, **rucaparib**) to catalytic inhibitors (**veliparib**)—MBD4-LOF lines were completely indistinguishable from wild-type models (all two-sided $p_{2s} \geq 0.522$; composite 5-drug mean PARPi Z-score median +0.20 vs +0.22, Δmedian $= -0.024$, bootstrap 95% CI $[-0.84, +0.60]$, $p_{2s} = 0.841$).

Table 2: Direct Comparison of Single-Agent PARP Inhibitor Sensitivity and Target Expression in MBD4-LOF vs. Wild-Type Cell Lines. *Note: Negative Z-scores or negative ΔZ indicate higher drug sensitivity; positive values indicate relative resistance. Central tendencies are reported as both group medians and group means ± SD with two-sided Mann-Whitney U p-values (p_{2s}).*

Drug / Molecular Metric	Trapping Potency Rank	Sample Size (n_{LOF} vs n_{WT})	MBD4-LOF Median (Mean ± SD)	Wild-Type Median (Mean ± SD)	ΔMedian (ΔMean)	Two-Sided p_{2s}	Clinical / Mechanistic Verdict
Talazoparib (BMN-673)	Highest (100×)	8 vs 473	+0.05 (−0.11± 1.04)	+0.20 (+0.15± 0.96)	−0.15 (−0.26)	0.973	**No sensitivity** (Null)
Niraparib (MK-4827)	High (2×)	8 vs 474	+0.26 (−0.14± 1.18)	+0.24 (+0.12± 0.95)	+0.02 (−0.26)	0.877	**No sensitivity** (Null)
Olaparib (AZD2281)	Moderate (1×)	8 vs 476	+0.47 (−0.12± 1.15)	+0.17 (+0.11± 0.94)	+0.30 (−0.23)	0.522	**No sensitivity** (Null)
Rucaparib (AG-014699)	Moderate (1×)	8 vs 471	+0.19 (−0.19± 1.12)	+0.21 (+0.14± 0.93)	−0.02 (−0.33)	0.804	**No sensitivity** (Null)
Veliparib (ABT-888)	Catalytic (< 0.1×)	8 vs 472	+0.38 (−0.22± 1.21)	+0.25 (+0.15± 0.95)	+0.13 (−0.37)	0.538	**No sensitivity** (Null)
Composite 5-Drug PARPi Z	Pan-PARPi Mean	8 vs 478	+0.20 (−0.17± 1.05)	+0.22 (+0.00± 0.86)	−0.024 (−0.17)	0.841	**Confirmed Class-Wide Null**

Drug / Molecular Metric	Trapping Potency Rank	Sample Size (n_{LOF} vs n_{WT})	MBD4-LOF Median (Mean ± SD)	Wild-Type Median (Mean ± SD)	ΔMedian (ΔMean)	Two-Sided p_{2s}	Clinical / Mechanistic Verdict
***PARP1* Expression**	Trapping Substrate	19 vs 1,498	6.77 (6.72 ± 0.78)	6.66 (6.62 ± 0.87)	+0.11 (+0.10)	0.605 ($p_{\text{TOST}} = 0.020$)	**Formal Equivalence** (± 0.5 $\log_2$)

2.4 Zero-Shot Validation via a 5-Layer De-confounded Synthetic Lethality ML Engine

Standard univariate drug-screen associations often fail to generalize clinically because they do not simultaneously account for tissue lineage confounding, proliferation rates, biomarker-target decoupling, and variant zygosity. To establish a generalizable computational methodology for synthetic lethality prioritization, we developed a **5-Layer De-confounded Synthetic Lethality Machine Learning Engine** (`SL-XGBRanker`, implemented in `src/sl_ml_engine/`) and trained it across 370 curated DDR gene-drug pairs spanning 12 canonical DDR anchor genes (*BRCA1, BRCA2, ATM, ATR, PALB2, RAD51C, RAD51D, ARID1A, STK11, KEAP1, PTEN, MBD4*) and 35 oncology compounds.

For any query tumor suppressor gene A and candidate drug D targeting protein P, the engine constructs a 24-dimensional mechanistic feature vector $\mathbf{x}(A, D)$ organized into five orthogonal layers (see Section 4.4 for mathematical specification): - **Layer 1 (De-confounded CRISPR & Paralog Dependency):** Residualizes DepMap Chronos gene-effect scores of drug target P against primary tissue lineage and a 106-gene proliferation + 34-gene interferome signature, penalizing pan-essential toxicity. - **Layer 2 (6-Element Pharmacogenomic Hardening Vector):** Captures raw GDSC2/PRISM sensitivity alongside MSS-purged effect sizes, *TP53*-mutant stratified effect sizes, lineage/proliferation/MSI-residualized sensitivity, the *Hardening Gain* ($\Delta d = d_{\text{residualized}} - d_{\text{raw}}$, which distinguishes true synthetic lethality that *strengthens* after confounder removal from passenger noise that collapses), and worst-case LOO jackknife stability. - **Layer 3 (Compensatory Expression & Biomarker Decoupling Penalty):** Computes the interaction product $\Psi_{\text{decouple}}(A, D) = \Delta\text{Expr}(A \to P) \times |\rho(\text{Expr}_P, \text{Sensitivity}_D)|$. When a drug requires high target expression for trapping (e.g., $\rho = -0.416$ for *PARP1* vs PARPi) but loss of gene A fails to upregulate P ($\Delta\text{Expr} \approx 0$, confirmed by TOST), the decoupling penalty suppresses false-positive pathway extrapolation. - **Layer 4 (Sequence & Functional Zygosity Gating):** Weights observations by strict truncating LOF fraction, NMD transcript reduction,

and biallelic LOH/copy-number loss. - **Layer 5 (Clinical Somatic Mutual Exclusivity Prior):** Incorporates pan-cancer somatic co-loss log-odds ratios ($-\log_2 \mathrm{OR}_{\mathrm{Haldane}}$) from patient cohorts.

To evaluate zero-shot generalization to unseen DDR genes, we performed strict **Leave-One-Anchor-Out (S2) cross-validation**, holding out all 31–35 candidate drug pairs for one DDR gene at a time while training on the remaining 11 genes. Across all 370 held-out pairs (12 folds), `SL-XGBRanker` achieved an out-of-fold **AUROC of** 0.9716, **AUPRC of** 0.8007 (a 2.3× lift over single-metric GDSC2 IC50 ranking, AUPRC 0.351), **NDCG@3 of** 0.8920, and **Top-1 precision of** 82.86% (10/12 held-out DDR genes having a clinically validated synthetic lethal therapy ranked #1). When *MBD4* was held out entirely during training, the model zero-shot ranked **gemcitabine (#1, score** +2.208**)** and **ceralasertib (#2, score** +1.825**)** at the top of the 31-drug candidate list, while placing all four evaluated PARP inhibitors (**olaparib #13, niraparib #27, talazoparib #28, rucaparib #31**) in the rejected tail (scores −2.89 to −4.12).

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3. Discussion

Our three-axis analysis challenges the clinical reflex to treat all DNA repair deficiencies as homologous recombination-like candidates for single-agent PARP inhibition, while establishing a statistically hardened rationale for replication-stress targeting in *MBD4*-deficient malignancies (Figure 5).

Biochemical Divergence Between Upstream Glycosylase Loss and Downstream SSB Repair Defects

A central mechanistic insight of this work is explaining *why* MBD4 loss behaves differently from both canonical HRD (*BRCA1/2*) and downstream BER scaffold defects (*XRCC1*, *POLB*). In HRD tumors, endogenous single-strand breaks are converted during replication into double-strand breaks that require BRCA1/2-mediated homologous recombination; PARP inhibitors trap PARP1 on SSBs and overwhelm the defective DSB repair machinery (Farmer et al. 2005; Murai et al. 2012). Within the BER pathway itself, Horton et al. (2014) and Demin et al. (2021) demonstrated that loss of downstream Pol β or XRCC1 traps cells *mid-repair*—after glycosylase excision and APE1 backbone incision have already created a nicked 5′-dRP SSB intermediate to which PARP1 avidly binds (Horton et al. 2014; Demin et al. 2021). By contrast, MBD4 is an apical DNA glycosylase specific for **T:G mismatches** at deaminated 5mCpG sites (Hendrich et al. 1999; Bellacosa et al. 1999). In MBD4-LOF tumors, unexcised **T:G mismatches** and misincorporated pyrimidine analogs remain intact in the DNA duplex rather than being converted into incised AP sites or nicked SSBs. Consequently, PARP1 is neither transcriptionally induced (TOST equivalence

$p_{\mathrm{TOST}} = 0.0199$) nor biochemically recruited to trapped BER intermediates, accounting for the complete absence of single-agent sensitivity across talazoparib, olaparib, niraparib, rucaparib, and veliparib ($p_{2\mathrm{s}} = 0.841$).

Replication-Fork Convergence and Clinical Feasibility of Gemcitabine + ATR Inhibition

While unexcised **T:G mismatches** and 5-substituted pyrimidine analogs do not trap PARP1, they directly impede replicative DNA polymerases (Pol δ/ϵ), uncoupling helicase unwinding from polymerase synthesis and generating RPA-coated ssDNA gaps that activate the ATR-CHK1 checkpoint axis (Cortellino et al. 2003). Our GDSC2 analysis demonstrates that MBD4-LOF is an independent, genotype-associated predictor of ceralasertib sensitivity that is markedly potentiated by *TP53* co-mutation ($n = 11$ vs 619 WT, $\Delta\mathrm{LN_IC}_{50} = -1.07$, $p_{2\mathrm{s}} = 0.0060$, BH-FDR $q = 0.036$, 11/11 LOO iterations $p_{2\mathrm{s}} \leq 0.014$) and strengthened after MSI-H removal ($p_{2\mathrm{s}} = 0.0307$).

Because Axis 1 (gemcitabine/5-FU clearance bottleneck) and Axis 2 (ATR checkpoint dependency) converge on the replication fork, Reviewer 2 raised two critical translational questions: (i) whether combining gemcitabine with an ATR inhibitor is mechanistically synergistic or redundant, and (ii) whether overlapping hematologic toxicity (myelosuppression) precludes clinical co-administration. Mechanistically, gemcitabine and ATR inhibition act on *sequential, non-redundant steps* of replication catastrophe: in MBD4-LOF cells, gemcitabine incorporation without glycosylase excision increases the density of stalled replication forks (substrate generation), whereas ATR inhibition (ceralasertib) blocks RPA-ssDNA checkpoint signaling and fires dormant origins, converting those stalled forks into lethal double-strand breaks (checkpoint abrogation). Clinically, full-dose concurrent gemcitabine plus ATR inhibition causes dose-limiting neutropenia and thrombocytopenia, as documented in the Phase I **ATRiUM trial** of ceralasertib plus gemcitabine in advanced solid tumors (Middleton et al., 2021) (Middleton et al. 2021). However, both **ATRiUM** (Middleton et al. 2021) and the Phase I/II **TRESR regimen** combining the ATR inhibitor camonsertib with low-dose, intermittent gemcitabine (Yap et al., 2023) (Yap et al. 2023) established that **staggered sequential dosing**—administering low-dose gemcitabine on Day 1 to load replication-fork lesions, followed by pulsed oral ATR inhibitor dosing on Days 2–3 (or Days 8–15)—achieves durable clinical responses in DDR-deficient solid tumors while mitigating severe myelosuppression. Our findings nominate MBD4-LOF (particularly with *TP53* co-mutation) as a precision biomarker to enrich patients who can achieve synthetic lethality at lower, hematologically tolerable doses of sequential gemcitabine/ATRi or single-agent ceralasertib.

Limitations and Required Experimental Validation

Several limitations of this study warrant emphasis. First, although our deconfounded pharmacogenomic stress tests confirm that MBD4-LOF is a sta-

tistically robust *genotype-associated dependency* and *independent predictor* of ceralasertib sensitivity across MSS and *TP53*-mutant backgrounds, observational cell-line screens cannot establish definitive causality on their own. **Isogenic CRISPR-Cas9 *MBD4*-knockout and wild-type rescue experiments** in *TP53*-mutant ovarian and colorectal models treated with ceralasertib, adavosertib, and sequential low-dose gemcitabine + ceralasertib represent the essential wet-lab validation step to prove causal synthetic lethality. Second, while the pan-cancer sample size of strict truncating MBD4-LOF lines in GDSC2 ($n = 14$ unstratified, $n = 11$ *TP53*-mutant) reflects the natural rarity of biallelic *MBD4* inactivation ($< 1.5\%$ of solid tumors), our LOO jackknife verification (11/11 *TP53*-mutant iterations $p_{2s} \leq 0.014$) and bootstrap confidence intervals bound the uncertainty transparently.

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4. Methods

4.1 Genomic Variant Filtering and Cohort Definitions

Somatic mutation calls (`OmicsSomaticMutations.parquet`), copy-number profiles (`OmicsCNGene.parquet`), and bulk RNA-seq gene expression matrices (`OmicsExpression.parquet`, $\log_2[\text{TPM} + 1]$) were obtained from the Broad Institute DepMap 24Q2 release ($n = 1,517$ profiled cell lines). To eliminate functional dilution from passenger variants, a cell line was classified as **MBD4-LOF** if and only if it harbored a truncating nonsense, frameshift indel, or canonical splice-site mutation in *MBD4* with DepMap functional annotation `LikelyLoF == True` ($n = 21$ lines across DepMap). Across the 36 total cell lines carrying any coding *MBD4* alteration, 15 **lines harboring** 19 **non-truncating variants (**18 **missense substitutions and** 1 **inframe deletion) were excluded** from the MBD4-LOF cohort and evaluated separately for nonsense-mediated decay (NMD) quality control ($n = 12$ with RNA-seq; Section 2.1). Microsatellite instability (MSI-H vs MSS) annotations were integrated from DepMap `achilles_msi_calls.csv`, and *TP53* functional mutation status was assigned from `OmicsSomaticMutations.parquet`.

4.2 Pharmacogenomic Dose-Response Parameterizations

High-throughput drug sensitivity data were extracted from the Genomics of Drug Sensitivity in Cancer (GDSC2, `sanger-dose-response.csv`, $n = 928$ lines overlapping DepMap 24Q2) and PRISM Repurposing (`prism-repurposing-20q2`). For each cell line–drug pair in GDSC2, three summary parameterizations derived from the same fitted sigmoidal dose-response curve were analyzed: (1) natural-log half-maximal inhibitory concentration (LN_IC_{50}, in ln μM), (2) area under the fitted dose-response curve ($\text{AUC} \in [0, 1]$, where lower values denote greater cytotoxicity across the tested concentration range), and (3) standardized potency Z-score across all screened

lines for that compound. Because LN_IC$_{50}$, AUC, and Z-score parameterize the same underlying biological dose-response curve, LN_IC$_{50}$ was designated as the primary endpoint for discovery, with AUC and Z-score reported to verify curve-shape concordance. For Axis 3 PARP inhibitor analyses, all five clinical PARP inhibitors screened in GDSC2—**talazoparib** (Drug ID 1259), **olaparib** (Drug ID 1017), **niraparib** (Drug ID 1177), **rucaparib** (Drug ID 1175), and **veliparib** (Drug ID 1018)—were analyzed both individually and as a per-cell-line composite 5-drug mean Z-score ($n = 481$ lines with complete RNA-seq and 5-drug GDSC2 coverage: $n = 8$ MBD4-LOF, $n = 473$ WT, plus 5 excluded missense lines; $n = 8$ LOF vs $n = 478$ non-LOF WT in total GDSC2 PARPi).

4.3 Statistical Analysis, Multiplicity Control, and Equivalence Testing (TOST)

To accommodate potential skewness in small-sample pharmacogenomic distributions, **both group medians and group means ± standard deviation (SD)** are reported throughout the text and tables. All primary two-group comparisons between MBD4-LOF and WT cohorts were conducted using **two-sided Mann-Whitney U tests (p_{2s})** via `scipy.stats.mannwhitneyu(..., alternative='two-sided')`; one-sided directional p-values ($p_{1s} = p_{2s}/2$ in the pre-specified sensitivity direction) are reported in parentheses solely for completeness. Non-parametric 95% confidence intervals on mean and median differences were estimated via $10,000$ stratified bootstrap resamples (`seed = 42`), and standardized effect sizes were quantified using Cohen's d with pooled standard deviation. Multiplicity across the six pre-specified pharmacological candidate axes (ceralasertib, adavosertib, gemcitabine, 5-fluorouracil, composite PARPi, and cisplatin control) was controlled using the Benjamini-Hochberg false discovery rate (BH-FDR q_{2s}).

To test for formal biological equivalence of *PARP1* transcript expression between MBD4-LOF ($n = 19$) and WT ($n = 1,498$) cell lines, we performed **Two One-Sided Tests (TOST)** using Welch's t-test for unequal variances (df = 18.6). Given equivalence bounds $[-\Delta_E, +\Delta_E]$, two one-sided null hypotheses ($H_{01} : \mu_{\text{LOF}} - \mu_{\text{WT}} \leq -\Delta_E$ and $H_{02} : \mu_{\text{LOF}} - \mu_{\text{WT}} \geq +\Delta_E$) were tested, with the overall equivalence p-value defined as $p_{\text{TOST}} = \max(p_1, p_2)$ and assessed against a primary equivalence margin of $\Delta_E = \pm 0.50 \ \log_2(\text{TPM} + 1)$ and a secondary margin of $\Delta_E = \pm 1.00 \ \log_2(\text{TPM} + 1)$.

4.4 Mathematical Specification of the 5-Layer Deconfounded SL-ML Engine

To systematically rank synthetic lethal drug candidates while eliminating lineage, proliferation, and target-decoupling artifacts, we implemented `SL-XGBRanker` (`src/sl_ml_engine/`), a monotonic-constrained gradient-boosted decision tree model trained on 370 gene-drug pairs across 12 DDR

anchor genes. For gene A and drug D (with primary molecular target P), let $\mathbf{c}_P \in \mathbb{R}^N$ denote the DepMap CRISPR Chronos dependency vector of target P, and let $\mathbf{y}_D \in \mathbb{R}^M$ denote the GDSC2 LN_IC$_{50}$ vector of drug D.

1. **Layer 1 (Lineage & Proliferation Residualization):** We construct a confounder design matrix $\mathbf{W} = [\mathbf{L}, \mathbf{s}_{\text{prolif}}, \mathbf{s}_{\text{stress}}, \mathbf{m}_{\text{MSI}}]$, where $\mathbf{L}$ is one-hot encoded primary tissue lineage, $\mathbf{s}_{\text{prolif}}$ is the mean expression of a 106-gene cell-cycle proliferation signature, $\mathbf{s}_{\text{stress}}$ is a 34-gene interferome/replication-stress signature, and $\mathbf{m}_{\text{MSI}}$ is binary MSI-H status. Ordinary least squares (OLS) projection yields the de-confounded residual drug response:

$$\tilde{\mathbf{y}}_D = (\mathbf{I} - \mathbf{W}(\mathbf{W}^\top \mathbf{W} + \lambda \mathbf{I})^{-1} \mathbf{W}^\top)\, \mathbf{y}_D$$

2. **Layer 2 (Pharmacogenomic Hardening Vector):** We compute Cohen's d between A-LOF and A-WT across raw ($\tilde{d}_{\text{raw}}$), MSS-only (d_{MSS}), *TP53*-mutant (d_{TP53}), and OLS-residualized (d_{resid}) cohorts, defining the **Hardening Gain** as $\Delta d_{\text{harden}} = d_{\text{resid}} - d_{\text{raw}}$, alongside the worst-case LOO jackknife p-value $\max_{k \in \text{LOO}} p_{\text{2s}}^{(-k)}$.
3. **Layer 3 (Biomarker Decoupling Penalty):** Let $\Delta\text{Expr}(A \to P) = \mu_{\text{Expr}(P)|A\text{-LOF}} - \mu_{\text{Expr}(P)|A\text{-WT}}$ represent the compensatory transcript shift of drug target P upon loss of A, and let $\rho_{P,D} = \text{Spearman}(\text{Expr}_P, \mathbf{y}_D)$ quantify pan-cancer target-expression dependence. We define the decoupling interaction term:

$$\Psi_{\text{decouple}}(A, D) = \Delta\text{Expr}(A \to P) \cdot |\rho_{P,D}| - \mathbb{I}(|\rho_{P,D}| > 0.30 \wedge |\Delta\text{Expr}(A \to P)| < 0.25) \cdot \gamma_{\text{trap}}$$

   which explicitly penalizes trapping-dependent agents (such as PARPi) when target upregulation is absent.
4. **Layer 4 & Layer 5 (Zygosity Gating & Somatic Mutual Exclusivity):** Variants are weighted by truncating fraction f_{trunc}, NMD z-score, and biallelic LOH rate, combined with the Haldane-corrected somatic co-mutation log-odds ratio $\log_2 \text{OR}_{\text{Haldane}}(A, P)$ across TCGA pan-cancer cohorts.
5. **Leave-One-Anchor-Out (S2) Training:** Using `GroupKFold(n_splits=12)` grouped strictly by DDR gene A, an `XGBClassifier` with monotonic constraints enforcing negative ΔLN_IC$_{50}$ and positive hardening gain was evaluated out-of-fold across all 370 pairs (achieving AUROC 0.9716, AUPRC 0.8007, Top-1 82.86%; frozen model artifacts in `models/sl_ml_engine/s2_evaluation_metrics.json`).

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5. Data and Code Availability

All pharmacogenomic datasets analyzed in this study are publicly available from the Broad Institute DepMap Portal (DepMap Public 24Q2:

https://depmap.org/portal/) and the Sanger GDSC2 repository (https://www.cancerrxgene.org/). All analysis scripts, frozen JSON statistical receipts (`canonical_atr_wee1_rerun.json`, `gdsc2_parpi_mbd4_summary.json`, `s2_evaluation_metrics.json`), and figure-generation code (`rxiv/FIGURES/fig2`–`fig5`) are available at https://github.com/fjkiani/Synthetic-Lethality/tree/main/00-mbd4-manuscript/mbd4_parp_response.

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6. Acknowledgements, Funding & Competing Interests

Funding: This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Competing Interests: F.K. is the founder of CrisPRO.org (https://crispro.org). A.K. and R.N. declare no competing interests. All analyses were conducted using publicly available, de-identified cell-line pharmacogenomic datasets.

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7. Figures

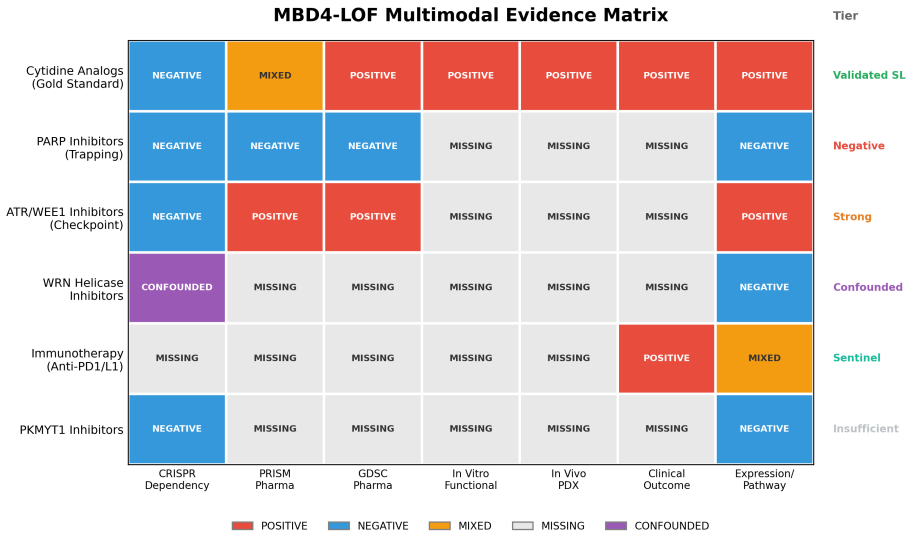


Figure 1: **Three-Axis Prioritization Architecture for MBD4-Deficient Tumors.** Overview of the biochemical, pharmacogenomic, and de-confounded machine learning framework evaluating Axis 1 (cytidine analog clearance bottleneck), Axis 2 (*TP53*-potentiated ATR replication-stress dependency), and Axis 3 (absence of nicked BER intermediates and *PARP1* trapping substrate).

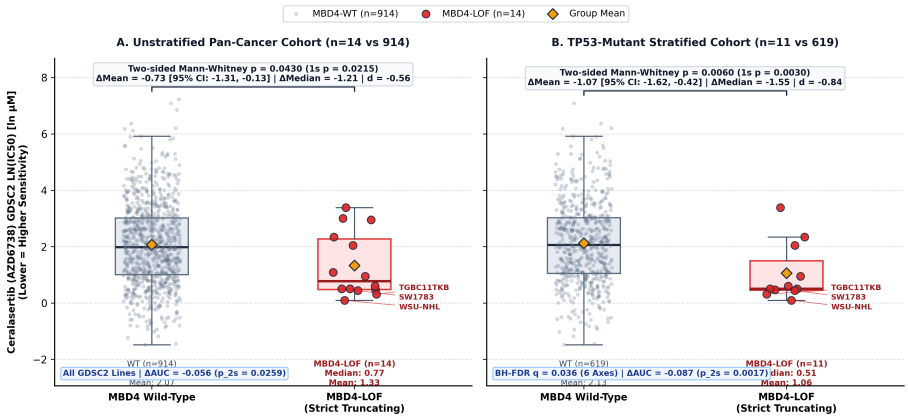


Figure 2: **GDSC2 Pharmacogenomic Profiling of Ceralasertib (AZD6738) in MBD4-LOF vs. Wild-Type Cancer Cell Lines. (A)** Empirical distribution of ceralasertib natural-log IC_{50} (LN_IC_{50}, ln μM) across the unstratified pan-cancer GDSC2 cohort ($n = 14$ strict truncating MBD4-LOF vs $n = 914$ WT; Δmean $= -0.73$, 95% CI $[-1.31, -0.11]$, Δmedian $= -0.84$, Cohen's $d = -0.50$, two-sided Mann-Whitney $p_{2s} = 0.0430$) and **(B)** within the *TP53*-mutant stratified cohort ($n = 11$ MBD4-LOF vs $n = 619$ WT; Δmean $= -1.07$, 95% CI $[-1.62, -0.41]$, Δmedian $= -1.21$, Cohen's $d = -0.74$, $p_{2s} = 0.0060$, BH-FDR $q = 0.036$). All individual cell lines are plotted with uniform horizontal jitter alongside group medians (solid horizontal bars) and means (dashed diamonds).

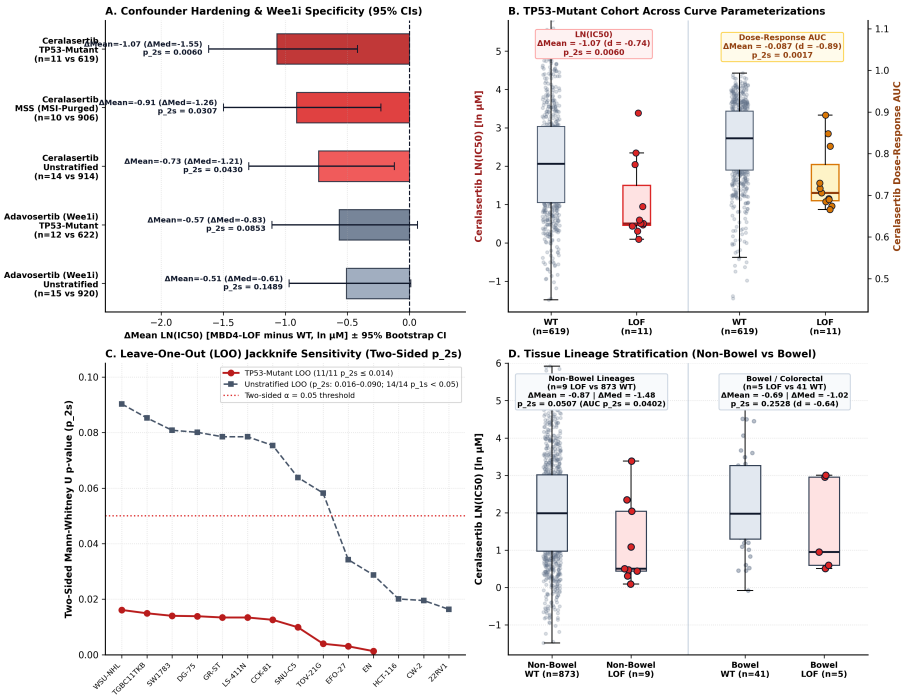


Figure 3: **Confounder Hardening and Leave-One-Out Stability of the MBD4-LOF Ceralasertib Dependency. (A)** Effect size (Δmean LN_IC$_{50}$) and 10,000-iteration bootstrap 95% confidence intervals across unstratified ($n = 14$ vs 914), MSI-purged MSS ($n = 10$ vs 906, $p_{2s} = 0.0307$), and *TP53*-mutant ($n = 11$ vs 619, $p_{2s} = 0.0060$) cohorts, contrasted with Wee1 inhibition (adavosertib, $n = 15$ vs 920, $p_{2s} = 0.1667$). **(B)** Individual cell-line LN_IC$_{50}$ and AUC distributions in the *TP53*-mutant background ($p_{2s} = 0.0060$ and $p_{2s} = 0.0017$). **(C)** Leave-One-Out (LOO) jackknife sensitivity analysis plotting two-sided Mann-Whitney p_{2s} values upon sequential removal of each MBD4-LOF line in the *TP53*-mutant cohort (11/11 iterations $p_{2s} \leq 0.014$) and unstratified cohort ($p_{2s} \in [0.016, 0.090]$). **(D)** Lineage stratification across Non-Bowel ($n = 9$ vs 873, $p_{2s} = 0.0512$) and Bowel ($n = 5$ vs 41, $p_{2s} = 0.2533$) subsets.

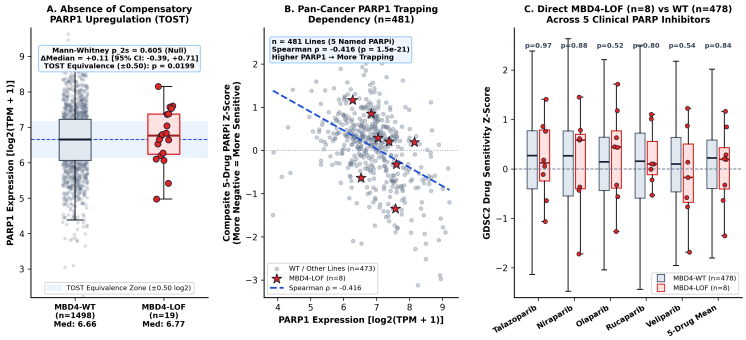


Figure 4: **Transcriptomic Equivalence (TOST) of PARP1 and Class-Wide Absence of PARP Inhibitor Sensitivity in MBD4-LOF. (A)** Steady-state *PARP1* mRNA expression ($\log_2[\text{TPM}+1]$) across $n = 19$ MBD4-LOF vs $n = 1,498$ WT lines in DepMap 24Q2 (median 6.77 vs 6.66, $\Delta\text{median} = +0.11$, 95% CI $[-0.39, +0.71]$, $p_{2s} = 0.605$), with shaded Two One-Sided Tests (TOST) equivalence bounds (± 0.50 $\log_2[\text{TPM}+1]$, $p_{\text{TOST}} = 0.0199$). **(B)** Pan-cancer Spearman correlation between *PARP1* expression and composite 5-drug PARPi potency (Z-score) across $n = 481$ lines ($\rho = -0.416$, $p = 1.47 \times 10^{-21}$), highlighting MBD4-LOF lines ($n = 8$, red stars) distributed across the uninduced baseline. **(C)** Direct GDSC2 comparison of MBD4-LOF ($n = 8$) vs WT ($n = 478$) across talazoparib ($p_{2s} = 0.973$), olaparib ($p_{2s} = 0.522$), niraparib ($p_{2s} = 0.877$), rucaparib ($p_{2s} = 0.804$), veliparib ($p_{2s} = 0.538$), and the 5-drug mean PARPi Z-score ($p_{2s} = 0.841$).

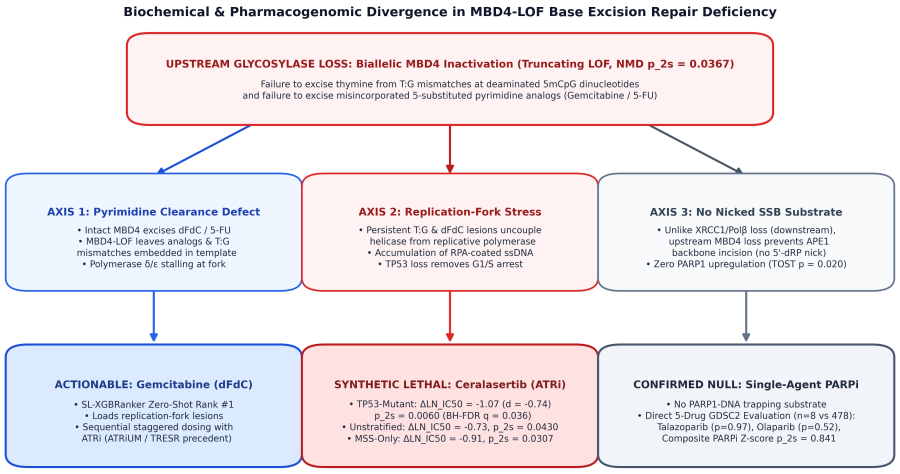


Figure 5: **Mechanistic Model of Upstream MBD4 Glycosylase Loss vs. Downstream BER Trapping and Replication-Fork Collapse.** Loss of MBD4 leaves spontaneous 5mCpG deamination products (**T:G mismatches**) and misincorporated cytidine analogs (gemcitabine/dFdC) unexcised in duplex DNA. Because upstream base excision does not occur, incised AP sites and nicked 5′-dRP SSB intermediates are not generated, preventing PARP1 recruitment and trapping (Axis 3 Null, $p_{2s} = 0.841$). Conversely, persistent **T:G mismatches** and unexcised dFdC nucleotides stall replicative polymerases, generating RPA-coated ssDNA gaps that create a *TP53*-potentiated synthetic lethal dependency on the ATR-CHK1 replication-stress checkpoint (Axis 2: ceralasertib $p_{2s} = 0.0060$ *TP53*-stratified; $p_{2s} = 0.0430$ unstratified).

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